FLAT REDUCTIONS WITH FAITHFUL FLATNESS IN DEGREE ONE GIVE RIG-SURJECTIVE FORMAL SPECTRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves the lemma in full, constructing the lift via the completed base change and a one-dimensional quotient embedded into a DVR

Editorial note. The label lemma-faithfully-flat-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- formal spectra of adic Noetherian rings and morphisms given by continuous ring maps
- rig-surjectivity tested by lifts along adic maps to complete discrete valuation rings
- the completed base change C = B completed tensor_A R and its reduction properties
- going-down for flat maps and the Jacobson complement-of-a-closed-point lemma
- existence of a one-dimensional local domain quotient embedding into a discrete valuation ring

Source reference. Original lemma 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Prerequisite pointer invoked at proof line(s) 5451; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN0: sources/stacks-chow/tags.txt, source lines 13423–13423. Prerequisite pointer invoked at proof line(s) 5429; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02IM: sources/stacks-chow/tags.txt, source lines 3305–3305. Prerequisite pointer invoked at proof line(s) 5448; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06FQ: sources/stacks-restricted/pinned-original-source.tex, source lines 426–426. Prerequisite pointer invoked at proof line(s) 5436; it names a standard fact of the same chapter that the counted argument uses, and

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every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to an ideal I. Let B be an I-adically complete A-algebra. If $A/I^n \to B/I^nB$ is of finite type and flat for all n and faithfully flat for $n = 1$, then $Spf(B) \to Spf(A)$ is rig-surjective.*

Proof. We will use without further mention that morphisms between formal spectra are given by continuous maps between the corresponding topological rings, see Formal Spaces, Lemma 0AN0. Let $\varphi : A \to R$ be a continuous map into a complete discrete valuation ring A. This implies that $\varphi(I) \subset \mathfrak{m}_R$. On the other hand, since we only need to produce the lift $\varphi' : B' \to R'$ in the case that φ corresponds to an adic morphism, we may assume that $\varphi(I) \neq 0$. Thus we may consider the base change $C = B\widehat{\otimes}_A R$, see Remark 06FQ for example. Then C is an $\mathfrak{m}_R$-adically complete R-algebra such that $C/\mathfrak{m}_R^n C$ is of finite type and flat over $R/\mathfrak{m}_R^n$ and such that $C/\mathfrak{m}_R C$ is nonzero. Pick any maximal ideal $\mathfrak{m} \subset C$ lying over $\mathfrak{m}_R$. By flatness (which implies going down) we see that $\mathrm{Spec}(C_\mathfrak{m}) \setminus V(\mathfrak{m}_R C_\mathfrak{m})$ is a nonempty open. Hence We can pick a prime $\mathfrak{q} \subset \mathfrak{m}$ such that $\mathfrak{q}$ defines a closed point of $\mathrm{Spec}(C_\mathfrak{m}) \setminus \{\mathfrak{m}\}$ and such that $\mathfrak{q} \notin V(IC_\mathfrak{m})$, see Properties, Lemma 02IM. Then $C/\mathfrak{q}$ is a dimension 1-local domain and we can find $C/\mathfrak{q} \subset R'$ with R' a discrete valuation ring (Algebra, Lemma 00PH). By construction $\mathfrak{m}_R R' \subset \mathfrak{m}_{R'}$ and we see that $C \to R'$ extends to a continuous map $C \to (R')^\wedge$ (in fact we can pick R' such that $R' = (R')^\wedge$ in our current situation but we do not need this). Since the completion of a discrete valuation ring is a discrete valuation ring, we see that the assumption gives a commutative diagram of rings

$$\begin{array}{ccccc} (R')^\wedge & \longleftarrow & C & \longleftarrow & B \\ \uparrow & & \uparrow & & \uparrow \\ R & \longleftarrow & R & \longleftarrow & A \end{array}$$

which gives the desired lift. □